

Conventional Sources

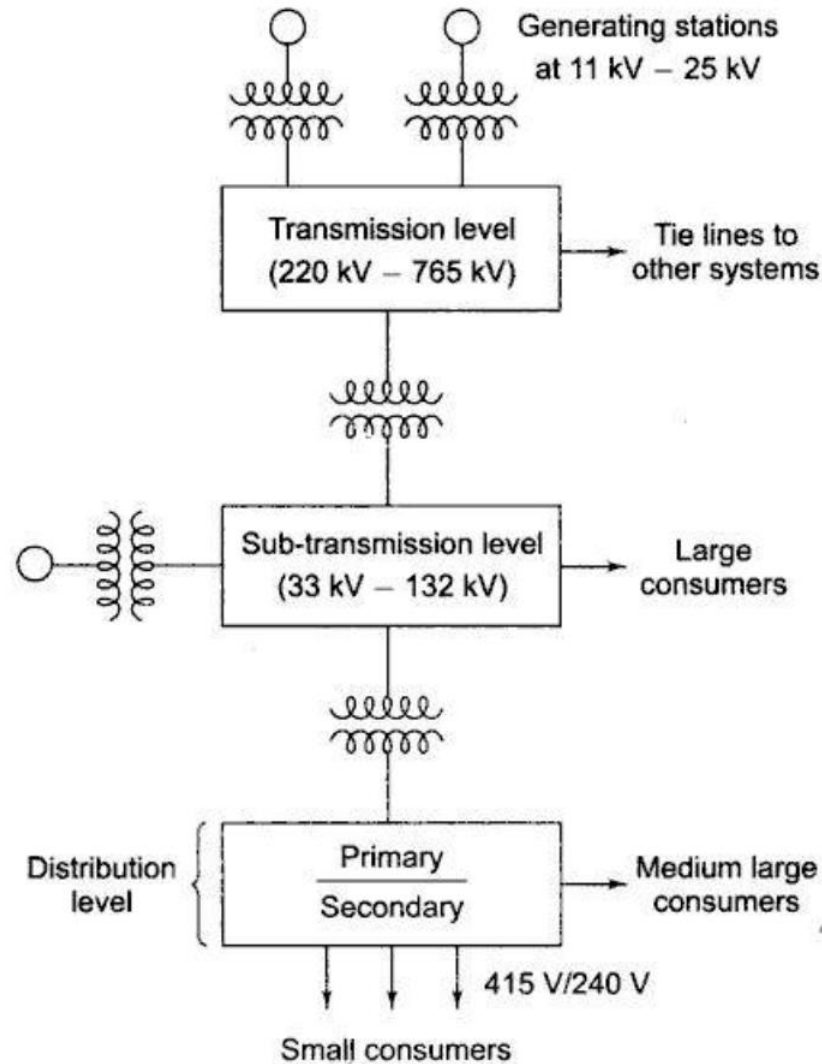
Dr. Sandeep J.

Assistant Professor

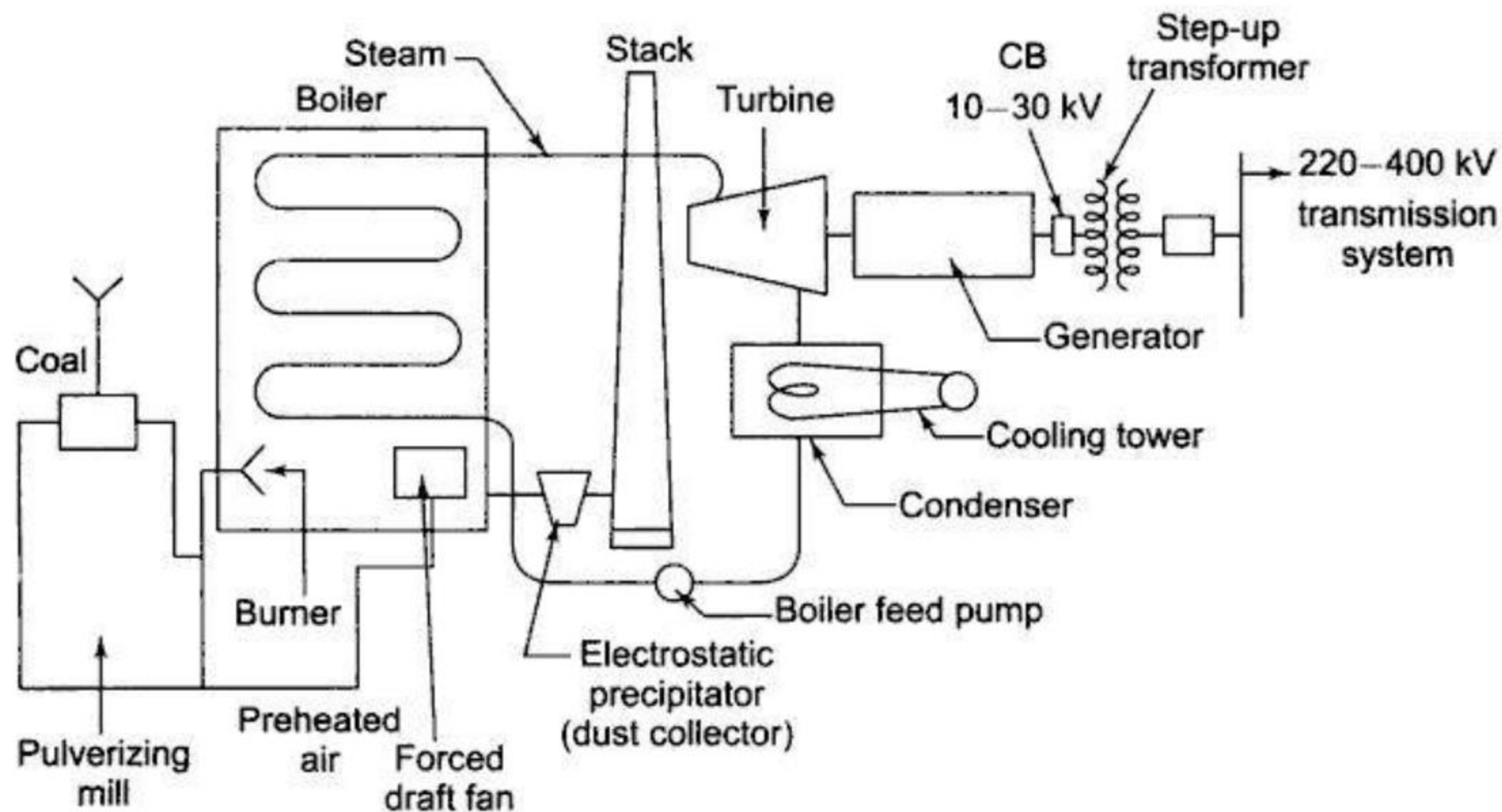
Department of Electrical and Electronics Engineering

Muthoot Institute of Technology and Science

Structure of Power System



Thermal Power Stations – Steam/Gas Based



- Chemical energy stored in coal is converted to heat energy – combustion of coal.
- Heat released produces steam inside the boiler at high pressure and temperature.
- While passing through the steam turbine, a part of the internal energy of steam is converted to mechanical energy.
- Turbine used is axial flow type – several such units on the same shaft.
- Steam turbine acts as the prime mover for the electric generator / alternator.
- Maximum efficiency of the overall conversion process is 40% only – can be improved by using the steam at highest possible pressure and temperature.

- **Coal and ash handling plant:** Coal is transported & stored in coal storage plant and delivered to coal handling plant. Coal burnt in boiler & ash produced is removed to ash handling plant and delivered to ash storage plant for disposal.
- **Boiler:** The heat of combustion of coal in the boiler is utilized to convert water into steam at high temperature and pressure.
- **Super heater:** steam produced in the boiler is wet and is passed through super heater where it is dried and superheated by flue gases on their way to chimney.
- **Economiser:** It's a feed water heater & derives heat from flue gases.
- **Air preheater:** It increases temperature of air supplied for coal burning by delivering heat from flue gases.

- Condenser

- The steam coming out of the turbine is condensed to water using a condenser and cooling tower arrangement
- The condensed water is very pure and recycled through the process.
- The boiler feed pump sends this water to the boiler through the economizer.

- Turbine and generator

- Dry and superheated steam from the super heater is fed to the steam turbine through main valve
- Heat energy of steam when passing through blades of turbine converted to mechanical energy.
- Steam turbines are used to rotate the alternator which converts mechanical energy to electrical energy.
- Electrical energy from alternator is delivered to the bus bars through transformer, circuit breaker and isolators.

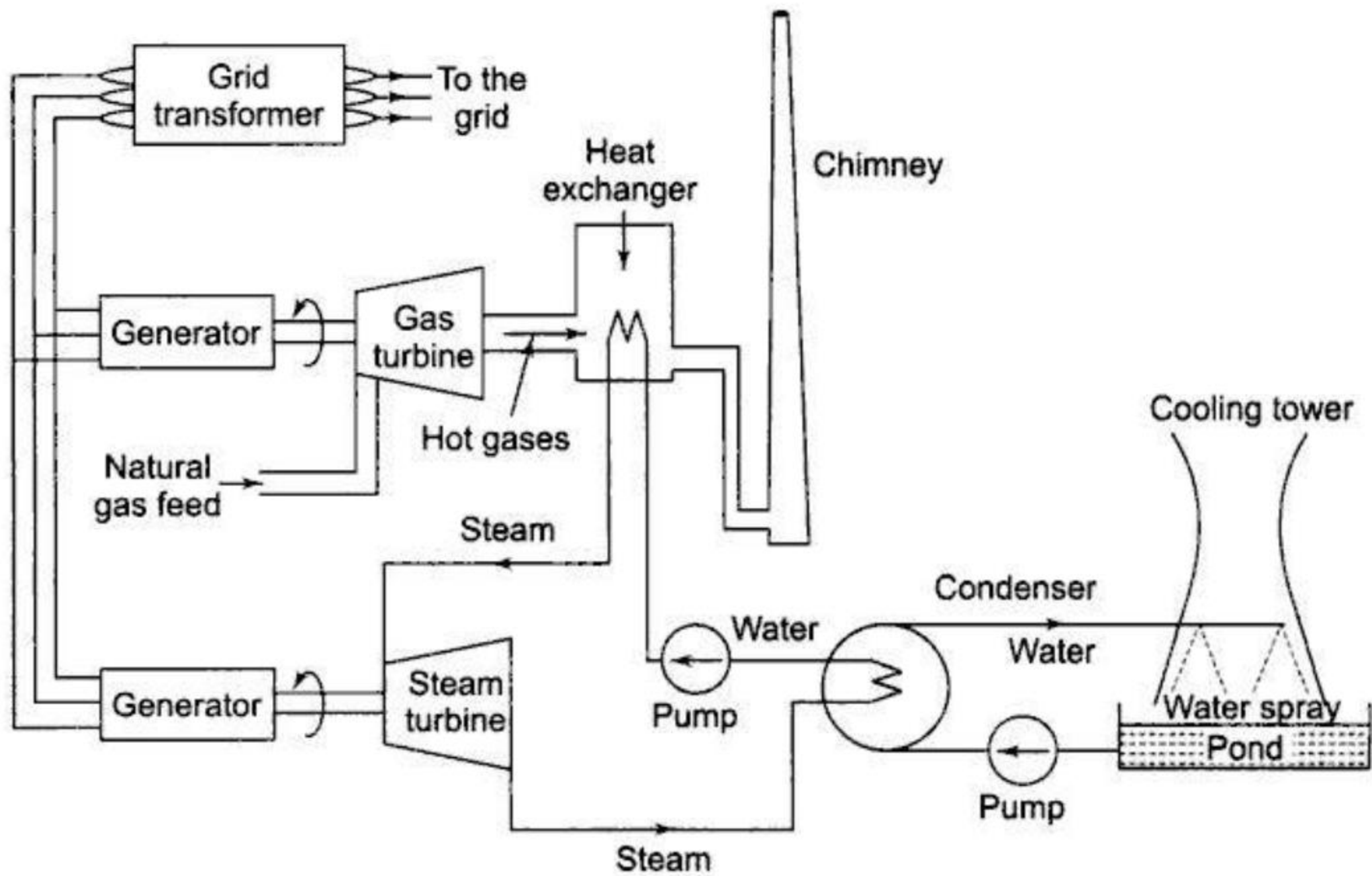
- Advantages:

- Fuel used is cheaper
- Less space required.
- Cheaper in initial cost.
- Cheaper in production cost.
- Can be installed at any place irrespective of existence of fuel.

- Disadvantages

- High maintenance and operating cost.
- Pollution of atmosphere – fumes and residues.
- Requirement of water in huge quantity.
- Handling of coal and disposal of ash is a difficult task.
- Long gestation period.

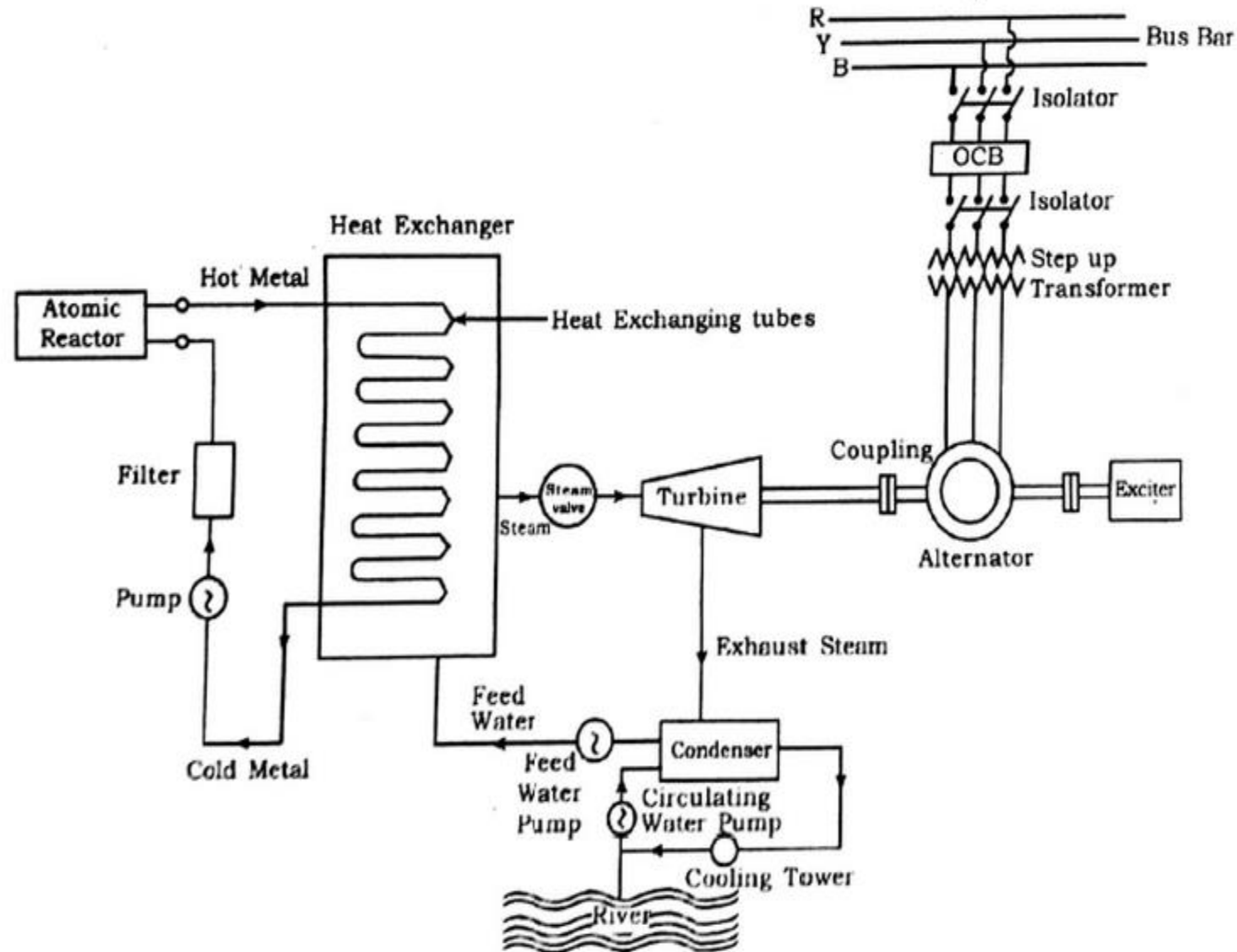
- Air and thermal pollution:
 - Air polluting agents (NO_x , CO , CO_2 , SO_x etc.) are emitted via the exhaust gases.
 - Thermal pollution due to rejected heat transferred from condenser to cooling water.
- Cogeneration – simultaneous generation of electricity and steam (or hot water) for industrial use or space heating – improves the efficiency – 65%.
- Gas Turbines:
 - Increased availability of natural gas (methane) – prime movers based on gas turbines has been developed – similar to those used in aircrafts.
 - Gas combustion generates high pressure and temperature – comparable efficiency to that of steam turbine.
 - Exhaust gas has high heat content which is used to raise steam – a conventional steam based generating system can be operated – Combined Cycle Gas Turbine (CCGT) plant.
 - Low CO_2 production when compared to coal based thermal plants.



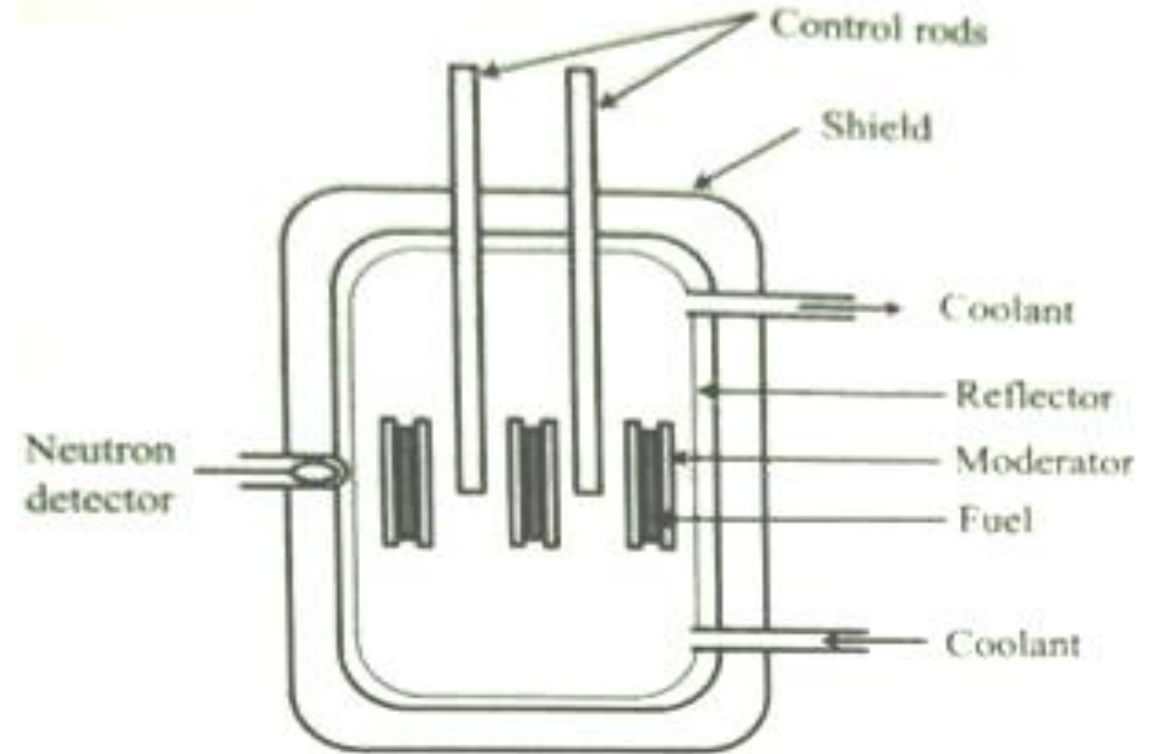
Nuclear Power Generation

- Nuclear reaction: Larger nucleus is broken to smaller fragments, thereby releasing considerable amount of binding energy – fission.
- For example, nucleus of uranium isotope undergoes fission when struck by a fast moving neutron.
- Nuclear reactor – an efficient source of energy:
 - The fission of a nuclear material is carried out in a nuclear reactor – small amount of fissile material produces large quantities of energy. (1g of uranium 235 produces 1 MW/day)
 - Heavy elements such as Uranium U 235 and Thorium Th 232 are subjected to fission in a reactor.
 - When atoms of nuclear fuels absorb free neutrons, they split into two or more smaller nuclei and some free neutrons.
 - Once fission of nuclear material is initiated, a chain reaction results – reactor core or moderator must slow down the moving neutrons.
 - Heat released by fission is removed by a gas or liquid coolant – used as the heat exchanger to raise steam.
 - If coolant is water, steam can be raised inside the reactor.

NUCLEAR POWER PLANTS (Cont'd..)



- Nuclear reactor – cylindrical stout pressure vessel – houses fuel rods, moderator (graphite) and control rods.
- Moderator rods encloses fuel rods- slows down the neutrons.
- Control rods (cadmium) absorbs neutrons – regulates fission.
- Coolant carries heat to the heat exchanger.



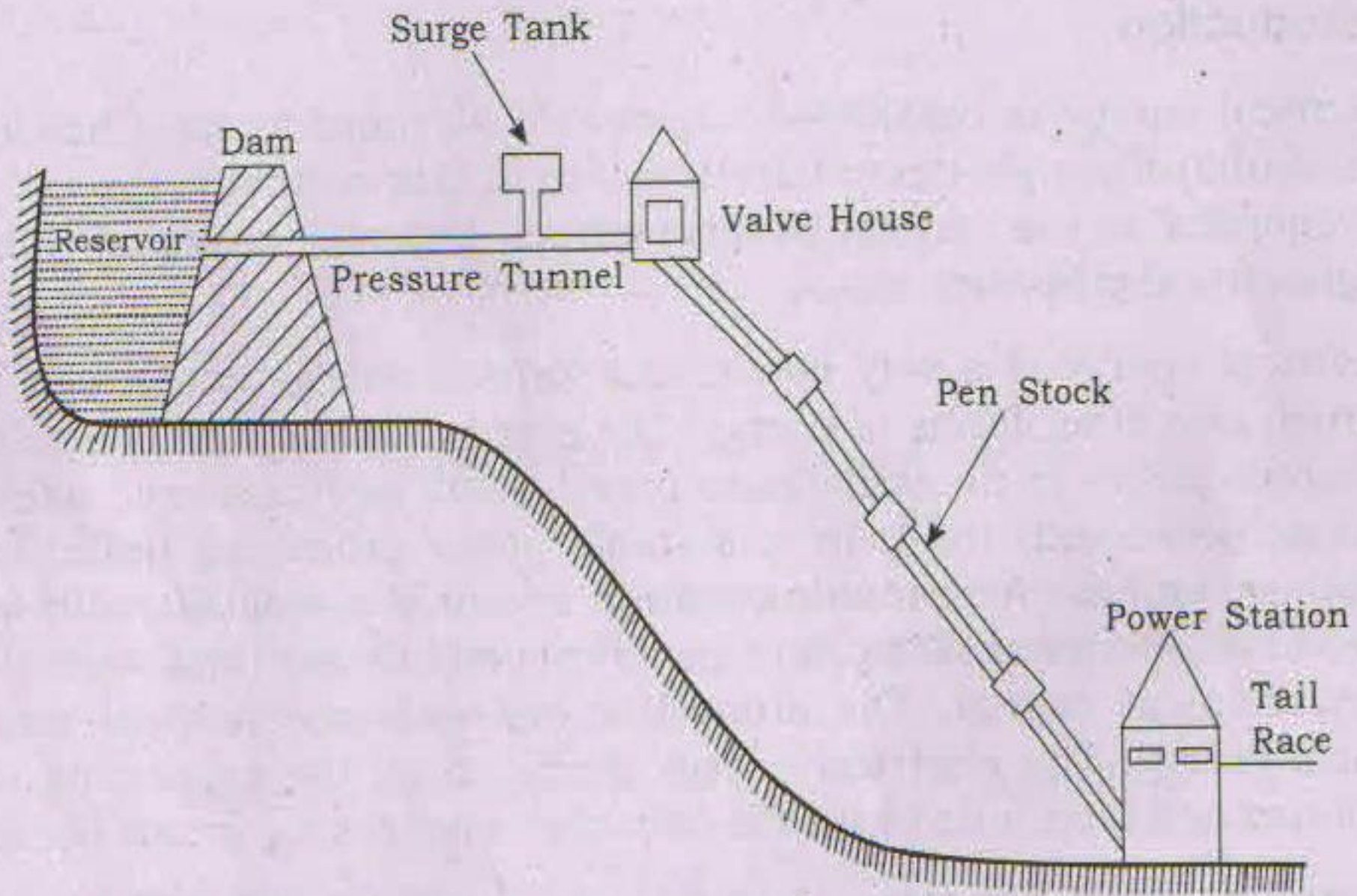
- Advantages:

- Amount of fuel required is small – saves transportation and storage cost.
- Lesser area required.
- Can be located near load centres.
- Economical if built in large quantity.
- Plenty of nuclear fuel deposits is available over the globe.

- Disadvantages

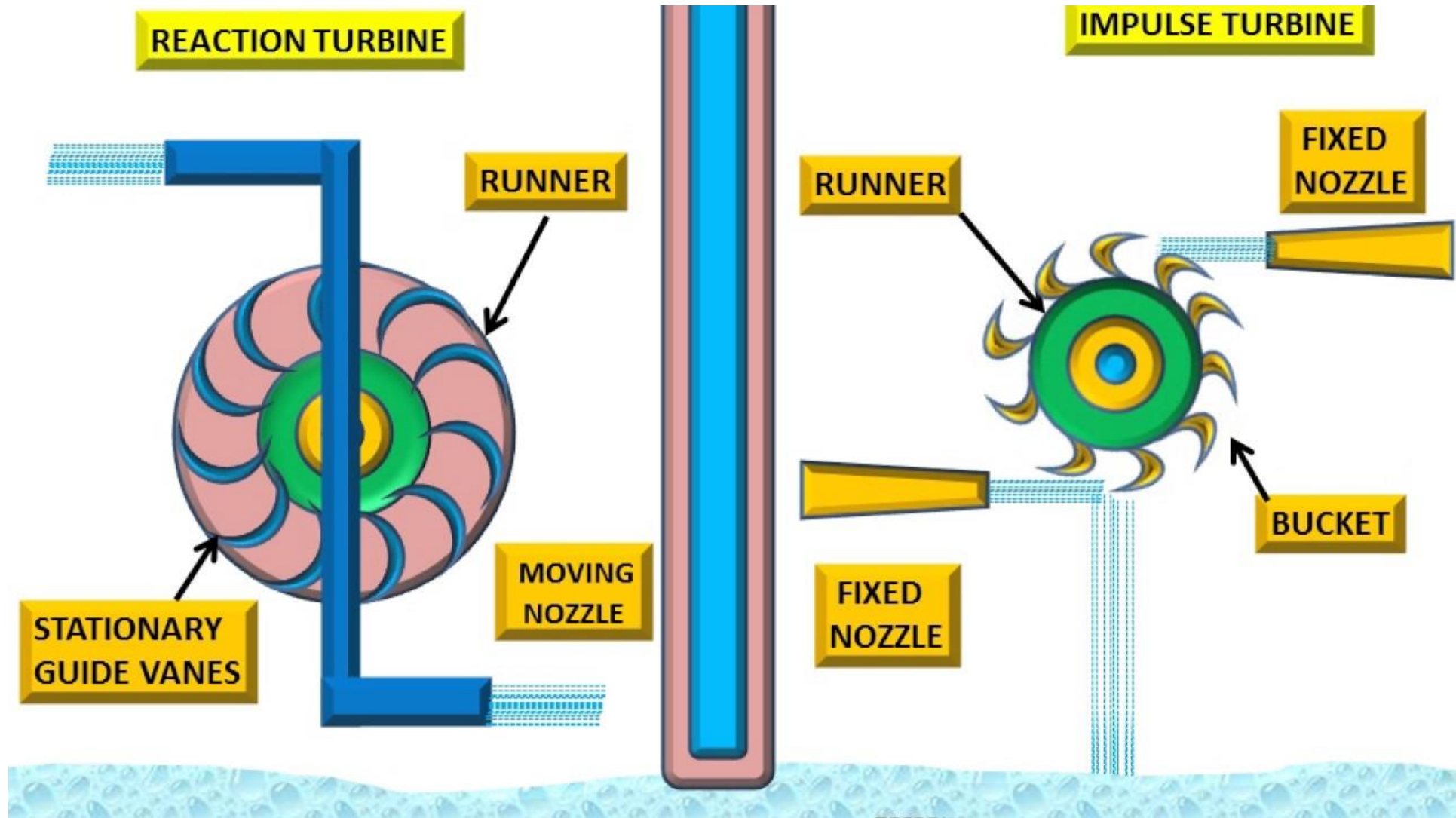
- High initial capital cost.
- Requires greater technical knowledge.
- Radio-active pollution.
- Fuel is costly.
- Disposal of by-products.

Hydro Electric Power Generation



- Converts the potential energy stored in water to electrical energy.
 - Water from a height is allowed to fall on the blades of a turbine through long pipes or tunnels called Penstock.
- Amount of power generated, $P = \rho Q g H$
- The dam is constructed across a river or lake and water from the catchment area collects at the back of the dam to form a reservoir. A pressure tunnel takes water to the valve house at the start of the penstock.
- The valve house contains main sluice valves and automatic isolating valves.
 - The former controls the water flow to the power house and the latter cuts off supply of water when the penstock bursts.
- From the valve house, water is taken to water turbine through penstock.
- The water turbine converts hydraulic energy into mechanical energy. The turbine drives the alternator which converts mechanical energy into electrical energy.
- A surge tank (open from top) is built just before the valve house.
 - Protects the penstock from bursting in case the turbine gates suddenly close due to electrical load being thrown off.

- Water turbines: Used to convert the energy of falling water into mechanical energy. The principal types of water turbines are :
 - Impulse turbines
 - Reaction turbines
- Impulse turbines: Used for high heads.
 - Turbine rotates due to water hitting the buckets of turbine.
 - The entire pressure of water is converted into kinetic energy in a nozzle and the velocity of the jet drives the wheel.
 - The example of this type of turbine is the Pelton wheel.
- Reaction turbines: Used for low and medium heads.
 - Turbine rotates due to force exerted by water when it flows inside the turbine.
 - In a reaction turbine, water enters the runner partly with pressure energy and partly with velocity head.
 - The important types of reaction turbines are :
 - Francis turbines
 - Kaplan turbines



- Advantages:

- No fuel is required – low operating cost.
- Highly reliable – operation and maintenance are cheap.
- No ash or smoke produced.
- Robust and longer life.
- Highly skilled engineers are required only at the time of construction.
- Irrigation and flood control.

- Disadvantages

- Large area required.
- Longer gestation period.
- Located away from load centres.
- Weather dependent.

Comparison:

Hydro Electric power plants	Thermal power plant	Nuclear power plant
Located near large reservoirs	Located where there is ample supply of water	Located away from populated areas
Initial cost is very high	Initial cost is lower than others	Initial cost is highest
Running cost is very less because no fuel	Running cost is highest	Higher than hydro power plant
Most efficient (85%)	Least efficient (25-35%)	More efficient (55%)
Weather depended	Limited fuel resource	Sufficient fuel resource
Practically no emission	High polluting emission	Clean but radiation problem
Maintenance cost is very low	Maintenance cost is very high	Maintenance cost is very high